

LIFEWS AgriShine(TM) GrowMeal(TM)

MASTER TEACHER GUIDE

Nursery | Primary 1-3 | Primary 4-6 | JSS 1-3 | SSS 1-3

Purpose	A common implementation, pedagogy, safety, assessment and quality-assurance standard for the complete GrowMeal school learning system.
Instructional cycle	10 garden weeks + Week 11 synthesis/remediation + Week 12 celebration and completion.
Evidence model	Know -> See -> Do -> Record -> Investigate -> Solve -> Assess -> Verify.
Resource families	Vocabulary, flashcards, demonstrations, practical cards, journals, missions, Ingredient Lab, Farm-to-Food, Food Smart Challenges, assessment, passports and lesson plans.

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1. PURPOSE AND SCOPE

What GrowMeal is	GrowMeal is the LIFEWS AgriShine school garden learning system. It treats the garden as a structured learning environment for food production, observation, practical competence, investigation, data use, food literacy, resource stewardship and age-appropriate problem solving.
Who uses this guide	School leaders, classroom teachers, garden coordinators, practical instructors, teaching assistants and other approved facilitators.
What this guide controls	The guide provides the common implementation standard across Nursery, Primary 1-3, Primary 4-6, JSS 1-3 and SSS 1-3. Level-specific resources provide the detailed learner tasks.

2. THE GROWMEAL LEARNING ARCHITECTURE

Core sequence	Learning Objective -> Vocabulary -> Flashcards -> Teacher Demonstration -> Practical Activity -> Journal -> Garden Mission -> Ingredient Lab / Farm-to-Food / Food Smart Challenge -> Assessment -> Food Smart Passport -> AgriShine Task.
Why the sequence matters	Learners first acquire language and concepts, then see a process demonstrated, practise it, record evidence, apply it to a challenge and demonstrate competence. The passport records verified achievement rather than simple attendance.
Teacher role	Select the appropriate resource, prepare the environment, model safe practice, ask productive questions, observe performance, protect data quality, provide feedback and verify evidence.

3. DEVELOPMENTAL PROGRESSION

Level	Learning progression	Expected evidence
Nursery	LOOK -> NAME -> COUNT -> MATCH -> CARE -> TELL	oral, visual, drawing, matching, counting and teacher-observed evidence
Primary 1-3	OBSERVE -> MEASURE -> COMPARE -> RECORD -> EXPLAIN	simple measurements, drawings, tables, sequences and short explanations
Primary 4-6	QUESTION -> MEASURE -> INVESTIGATE -> ANALYZE -> EXPLAIN	fair tests, repeated measurements, graphs, calculations, limitations and evidence-based conclusions
JSS 1-3	DEFINE -> HYPOTHESIZE -> EXPERIMENT -> QUANTIFY -> ANALYZE -> RECOMMEND	traceable datasets, replicated investigations, quantitative analysis, uncertainty and management recommendations
SSS 1-3	FRAME -> DESIGN -> INSTRUMENT -> MODEL -> VALIDATE -> EVALUATE -> OPTIMIZE	research protocols, instrumentation/QC, models, validation, trade-offs, design decisions and optimization

Progression is cumulative. Higher levels retain observation, care and practical competence while adding stronger measurement, experimental design, data analysis, systems reasoning and decision quality.

4. THE 12-WEEK INSTRUCTIONAL CYCLE

Weeks 1-10	Teach one level-specific GrowMeal garden per week. Use the garden as the anchor for vocabulary, demonstration, practical work, journal evidence, mission work and assessment.
Week 11	Synthesis, remediation and portfolio review. Revisit competencies that are not secure, complete missing evidence and moderate assessment records.
Week 12	GrowMeal celebration and demonstration. Learners communicate what they grew, observed, measured, investigated and solved. Complete passport verification and next-level goals.
Scheduling	Schools may divide a weekly lesson into classroom preparation and garden practical time. Preserve the learning sequence even when the timetable requires more than one session.

5. BEFORE EVERY PRACTICAL LESSON

Site check	Inspect beds, paths, water points, structures and learning stations before learners enter. Remove hazards and restrict access to unsafe areas.
Materials	Prepare only the tools and samples required. Count items before and after use. Provide low-cost DIY substitutes when these do not weaken safety or measurement validity.
Learner grouping	Assign small groups with clear roles such as observer, measurer, recorder, materials monitor or reporter. Rotate roles so competence is distributed.
Evidence plan	Decide before the lesson what evidence each learner or group must produce and where it will be recorded.
Contingency	Prepare a classroom alternative using samples, photographs, process cards, existing datasets or journal analysis when outdoor work cannot proceed.

6. SAFETY, SAFEGUARDING AND HYGIENE

Supervision	Learners work only within teacher-approved zones and tasks. Age, maturity, accessibility needs and school policy determine which tools a learner may use.
Tools and equipment	Teachers control knives, heat, electrical equipment, chemicals, powered machinery and other higher-risk equipment unless a formal school protocol specifically authorizes age-appropriate supervised use.
Food handling	No tasting is required. Use potable water for food-contact washing, clean food-grade containers, handwashing procedures and teacher-approved produce. Do not use unknown plants or produce with uncertain contamination or treatment history.
Allergies and dietary needs	Check school records and procedures before any food-contact activity. Ingredient observation does not require consumption.
Garden inputs	Follow product labels and school procedures for fertilizers, pest-management products and amendments. Learners should not handle hazardous inputs.
Incident response	Stop the activity, secure the area, follow school first-aid/reporting procedures and document any change needed before the activity is repeated.

7. TEACHING WITH VOCABULARY AND FLASHCARDS

Vocabulary	Introduce only the terms needed for the current experience. Learners should encounter the term in a real object, process, measurement or decision wherever possible.
Flashcards	Use flashcards to activate prior knowledge, identify objects/processes, rehearse terminology and check understanding. They are not a substitute for practical experience.
Language progression	Nursery emphasizes naming and oral language. Primary adds written descriptions and measurement language. JSS/SSS require increasingly precise scientific, systems and engineering terminology.

8. TEACHER DEMONSTRATION STANDARD

Demonstrate first	Show the procedure before independent learner practice when the task involves a new tool, measurement, sampling method or safety requirement.
Model thinking	State what you are checking, why a step matters and what would make the evidence unreliable.
Check understanding	Ask learners to explain or demonstrate the critical step before releasing groups to work.
Close the demonstration	Restate the success criterion, evidence requirement, cleanup procedure and stop signal.

9. PRACTICAL ACTIVITY IMPLEMENTATION

Purpose	Practical cards convert concepts into observable competence. The teacher should assess what learners actually do, not only what they can describe.
DIY materials	Low-cost materials are encouraged when clean, safe, sufficiently durable and appropriate to the measurement or learning objective.
Differentiation	Adjust quantity, complexity, recording format, group role, time and scaffolding without removing the core competence.
Troubleshooting	If a procedure fails, preserve the evidence. A failed germination, blocked dripper or inconsistent measurement can become an investigation when handled safely.

10. JOURNALS AND EVIDENCE

Evidence ownership	Each learner should have identifiable evidence appropriate to the level. Group data may be shared, but individual interpretation should be visible when developmentally appropriate.
Data integrity	Do not invent missing measurements. Mark missing data, explain anomalies and distinguish observation from interpretation.
Feedback	Comment on evidence quality, method, reasoning and next steps. Avoid rewarding decorative presentation more than scientific or practical competence.
Portfolio use	Select representative journal pages, mission evidence and assessment records for Week 11 review and passport verification.

11. GARDEN MISSIONS AND PROBLEM SOLVING

Mission purpose	Garden Missions ask learners to use what they know to complete a practical challenge rather than repeat a memorized fact.
Teacher support	Clarify constraints and safety, but do not solve the mission for the learner. Use prompts that return attention to evidence.
Success	Judge both the result and the reasoning/process. At higher levels, require criteria, uncertainty and a defensible recommendation.

12. INGREDIENT LAB

Purpose	Connect plants to recognizable ingredients through plant-part identification, structure, measurement, quality, post-harvest change and investigation.
Boundary	The Ingredient Lab is not a medical-claims program and does not require tasting.
Higher-level work	JSS and SSS may use hypotheses, replication, process mapping, quality criteria, validation and optimization while maintaining school-scale safety controls.

13. FARM-TO-FOOD LEARNING

Purpose	Trace the learning chain from production through harvest, handling, sorting, storage/preparation pathways, food literacy and waste reduction.
Key questions	Where did this food originate? Which plant part is used? What protects quality? Where can loss occur? What information supports a responsible choice?
Waste	Treat avoidable loss and organic residuals as learning opportunities for measurement and resource-management thinking, subject to hygiene rules.

14. FOOD SMART CHALLENGES

Purpose	Use real-world scenarios involving ingredients, food journeys, labels, meals, water, waste, seasonality, storage and integrated food-system decisions.
Evidence standard	Nursery may answer through matching, drawing and talk. Primary increasingly compares evidence. JSS/SSS should quantify, evaluate uncertainty and defend decisions.
Claims	Do not teach unverified therapeutic or medical claims. Keep food-literacy decisions grounded in observable information, labels, school guidance and evidence.

15. ASSESSMENT AND MODERATION

Four dimensions	Assess knowledge, practical competence, investigation/data skills and mission-based problem solving.
Rubric	For open/practical evidence use the established 0-3 scale: 3 Secure/Proficient, 2 Developing, 1 Beginning/Emerging, 0 Not demonstrated.
Multiple evidence sources	Use direct observation, journal work, practical outputs, mission evidence, data products and assessment items. Do not infer practical competence from a written quiz alone.
Moderation	Teachers should compare samples of learner work and discuss scoring decisions, especially near achievement-band boundaries.
Reassessment	After feedback and additional practice, learners may demonstrate the competence again according to school policy.

16. FOOD SMART PASSPORT

Purpose	The passport is a learner-owned achievement record linking gardens, practicals, journals, missions, Ingredient Lab, Farm-to-Food, Food Smart Challenges and assessment.
Verification	Teacher initials, stamps or signatures should represent demonstrated participation or competence, not attendance alone.
Completion	Use Week 11 evidence review before signing level completion. Record next-level goals where competence remains developing.

17. INCLUSION AND DIFFERENTIATION

Access	Plan pathways, work surfaces, tool choices, visual supports, communication methods and group roles so learners can participate meaningfully.
Scaffolding	Break complex tasks into steps, provide worked examples or templates, pair learners strategically and reduce recording load when the learning target is practical rather than writing.
Extension	Increase independence, sample size, variables, analysis, design constraints or communication demands for learners ready for additional challenge.
Fairness	Different supports may be used while the core learning objective and evidence standard remain clear.

18. GARDEN MANAGEMENT AS PEDAGOGY

Routine care	Watering, checking plants, cleaning tools, recording conditions and maintaining paths can become structured learning when linked to an objective and evidence.
Continuity	Keep garden records so classes can learn from previous cohorts and seasonal changes.
Failure as data	Crop loss, pest damage, water problems and design limitations should be documented and investigated rather than hidden, provided the site remains safe.

19. RECORD KEEPING AND QUALITY ASSURANCE

Minimum records	Keep lesson dates, attendance where required, selected resource IDs, assessment records, passport verification, incidents and relevant garden maintenance notes.
Traceability	At JSS/SSS, datasets should identify date, location/system, sample or treatment, units and responsible group/learner where practical.
Resource control	Use the unique PA, TD, GM, AQ, IL, FF and FSC identifiers when recording activities so evidence can be traced back to the instructional resource.
Review cycle	At the end of each term, review learner outcomes, garden performance, material shortages, safety observations and resource revisions.

20. SCHOOL LEADERSHIP IMPLEMENTATION

Roles	Assign a GrowMeal lead, participating teachers and responsibility for garden maintenance, materials, records and safety escalation.
Timetable	Protect regular practical time. A garden program becomes inconsistent when practical work is repeatedly displaced by classroom-only teaching.
Procurement	Prioritize safe basic tools, water access, durable learning materials, measurement equipment appropriate to the level and replacement consumables.
Community engagement	Use approved parents, local growers, technical professionals and food-system practitioners as learning contributors when safeguarding and school procedures are met.

21. TERM-END REVIEW

Learner outcomes	Which competencies became secure? Which remain developing? What evidence supports the judgment?
Instruction	Which demonstrations, practicals, missions and challenges produced the strongest learning? Which require revision or more time?
Garden system	What changed in productivity, maintenance, water use, infrastructure or learner access?
Next term	Set specific actions for teaching, garden maintenance, materials, assessment moderation and learner support.

22. MASTER IMPLEMENTATION CHECKLIST

Before term	Confirm gardens, timetable, staff roles, learner resources, safety procedures, materials and baseline garden condition.
Each week	Prepare objective and vocabulary; select resources; inspect site; demonstrate; facilitate practical; collect evidence; assess; update passport; complete cleanup/maintenance.
Week 11	Moderate evidence, remediate gaps, complete portfolios and verify passport records.
Week 12	Run the learning showcase, complete certification, capture reflections and set next-level/next-term goals.
After term	Archive records, review garden condition, replenish materials and document revisions for the next cycle.

APPENDIX A - WEEKLY TEACHER PLANNING TEMPLATE

Level / class	_____
Week / garden	_____
Learning objective	_____
Vocabulary	_____
Resource IDs	TD-____ PA-____ GM-____ AQ-____ IL-____ FF-____ FSC-____
Materials / DIY	_____
Evidence required	_____
Safety check	_____
Differentiation	_____
AgriShine task	_____

Post-lesson review:

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APPENDIX B - PRACTICAL OBSERVATION RECORD

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APPENDIX C - TERM QUALITY REVIEW

Learner outcomes	Which knowledge, practical, data/investigation and mission competencies were strongest/weakest?
Garden condition	What worked, failed or needs maintenance?
Safety	What incidents, near misses or procedural improvements were identified?
Resources	Which materials or cards need replacement, clarification or adaptation?
Assessment	Were scoring decisions consistent? Which evidence samples should be moderated?
Next actions	List responsible person, action and target date for each improvement.

GrowMeal(TM) implementation principle: the garden is not an occasional demonstration space. It is a managed learning system in which learners progressively know, do, investigate, solve and communicate.